Personal Statement

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UK Global Talent Visa — Digital Technology (Exceptional Promise)

I am an independent researcher and systems engineer based in Derby, United Kingdom. Over the past three years—without institutional affiliation, funding, or supervision—I have designed and built a blockchain-native compliance platform spanning 17 containerised microservices and 18 smart contracts, trained machine-learning models on 2.64 million Ethereum transactions, produced three original research papers establishing a unified mathematical theory of instability detection across finance, energy infrastructure, and fluid dynamics, filed a UK patent application covering 25 claims across 13 invention groups, published on TechRxiv and Zenodo, and open-sourced developer tools in TypeScript, Python, and Dart. I am applying under the Exceptional Promise route because my work demonstrates the capacity to become a leading figure in the UK's applied mathematics, financial technology, and machine learning sectors.

1. What I have built.

My central engineering project is **AMTTP**—the Anti-Money Laundering Transaction Trust Protocol. It addresses a problem that no existing commercial product has solved: enforcing regulatory compliance *before* a blockchain transaction reaches finality, rather than generating alerts after the fact. Chainalysis, Elliptic, and TRM Labs all operate post-settlement; AMTTP intervenes at the protocol level within the approximately 12-second Ethereum block confirmation window.

Architecture. The system spans four layers. At the outermost layer sit open-source client SDKs in TypeScript (19+ service endpoints covering risk, KYC, transactions, disputes, reputation, sanctions, GeoRisk, MEV protection, and bulk operations for up to 10,000 transactions) and Python (17 matching services, full async API). The compliance orchestration layer comprises 7 parallel microservices—ML risk scoring, graph-based relationship analysis, policy engine, sanctions screening, continuous monitoring, geographic risk, and an explainability module—coordinated through a central gateway with NGINX TLS termination, RS256 JWT validation, per-key rate limiting, and Ed25519 oracle-signed responses with 90-day rotation. The ML pipeline (described below) supplies risk scores to the orchestrator. Finally, the smart contract layer enforces on-chain compliance through 18 Solidity contracts deployed on Ethereum Sepolia, including risk-tiered policy enforcement (Approve / Review / Escrow / Block), Kleros-integrated decentralised dispute resolution, LayerZero cross-chain policy synchronisation, Gnosis Safe and Biconomy account-abstraction modules, and Compliance NFTs. The full infrastructure totals 17 containerised microservices—each with health checks, circuit breaker patterns, and

Prometheus / Grafana monitoring—orchestrated through Docker Compose.

ML pipeline. The production fraud-detection model uses a Composite Teacher architecture (40% XGBoost, 30% FATF pattern detectors, 30% graph structural properties) that generates pseudo-labels across 2.64 million Ethereum mainnet transactions sourced from Google BigQuery. A stacked ensemble of XGBoost (500 estimators) and LightGBM (500 estimators) with a logistic regression meta-learner achieves a production ROC-AUC of **0.9879**, PR-AUC of **0.9715**, and F1 of **0.9012** with 21 features and **zero training-serving skew**. An earlier v2 student pipeline integrating BetaVAE (64 latent dimensions), GraphSAGE (3 layers), and GATv2 (4 attention heads) reaches ROC-AUC 0.9229. Achieving zero skew was itself a significant engineering challenge: v2 models trained on 71 features but served with only 5 (66 features hardcoded to zero); diagnosing and correcting this lifted ROC-AUC from 0.64 to 0.99 in a single iteration.

zkNAF. I also built **zkNAF**—a zero-knowledge Non-disclosing Anti-Fraud framework using Groth16 ZK-SNARK proofs on the BN254 elliptic curve. zkNAF provides four proof types: sanctions non-membership (Merkle non-membership without revealing address or list), risk-range proofs (ML score within acceptable bounds without exposing the exact value), KYC verification (credential completeness without identity disclosure), and combined transaction compliance. This resolves the tension between GDPR's data minimisation requirements and AML transparency obligations at the cryptographic level, with dedicated on-chain verifier contracts routed through a single `ZkNAFVerifierRouter`.

Consumer application. To make pre-settlement compliance accessible to end users, I built a cross-platform Flutter consumer application targeting six platforms (iOS, Android, Web, Windows, macOS, Linux) from a single Dart codebase. The app includes a wallet dashboard, pre-transfer trust check with traffic-light interstitial, transaction history, dispute filing, and MetaMask wallet connection. It communicates with the compliance backend through a hybrid Flutter / Next.js architecture using a bidirectional JavaScript Channel bridge. A separate Next.js 15 "War Room" dashboard serves institutional compliance analysts.

Security architecture. Security spans every layer: EIP-712 typed signing for request integrity, server-enforced nonce + TTL (300 s) replay protection via Redis, HMAC-SHA256 webhook authentication, a 6-tier RBAC hierarchy, and a UI Integrity Service that monitors DOM mutations and verifies per-component hashes to prevent address-swap attacks of the kind used in the 2025 Bybit exploit.

The AMTTP architecture is described in my published paper: *Olusegun Odeyemi, "AMTTP: A Four-Layer Architecture for Deterministic Compliance Enforcement in Institutional DeFi," TechRxiv, February 27, 2026*, DOI: 10.36227/techrxiv.177220113.33816607/v1.

2. What I have discovered.

Alongside the engineering work, I have pursued a research programme that began with a practical question—why do fraud detectors miss what they are designed to find?—and evolved into a unified mathematical theory of instability in complex systems, with results spanning machine learning, systemic financial risk, electrical grid failure, algorithmic stablecoin collapse, and the regularity of the 3D Navier-Stokes equations.

Paper 1: Blind Spot Decomposition Theory (BSDT). My first paper provides a mathematical framework for characterising missed fraud. I decompose detection failures into four near-orthogonal components—Camouflage (δ_C), Feature Gap (δ_G), Activity Anomaly (δ_A), and Temporal Novelty (δ_T)—and prove that these account for over 95% of explainable variance in missed-fraud indicators (component correlations $|\rho| < 0.3$ for >80% of pairs; NMI < 0.10 for >90% of pairs). The correction operators recover recall by up to **84.4 percentage points** without retraining or access to model internals. Validated across 4 model architectures (XGBoost, LightGBM, Random Forest, Logistic Regression), 7 datasets from 6 domains (Bitcoin / Elliptic with 203,769 transactions, Ethereum addresses, IEEE-CIS credit card, NASA Shuttle telemetry, mammography, pen-digit recognition, e-commerce card-not-present), and over 36 model–dataset combinations, the best variant (F2_QuadSurf) achieves mean F1 of 0.787 with 88.1% false positive reduction. The paper also proves formal safety, boundedness, and monotonicity guarantees for the correction formula and introduces three self-calibrating weight estimation methods (mutual information, Fisher, Bayesian online). Published on Zenodo (DOI: 10.5281/zenodo.18870589); target journal: **IEEE TKDE**.

Paper 2: Universal Deviation Law (UDL). My second paper tackles anomaly detection from first principles. I introduce a multi-spectrum tensor framework with 14 composable operators spanning five mathematical families—statistical, dynamical, spectral, geometric, and exponential—each designed to capture a distinct axis of deviation. A 14×14 correlation audit confirms that only 1 pair out of 91 is redundant (Wavelet↔Phase, $\rho = 0.92$): every other operator adds genuinely new information. The best configuration achieves mean AUC of **0.972** and 91% coverage across five benchmarks—categorically outperforming Isolation Forest (0.788), LOF (0.632), and the current state-of-the-art ECOD (0.921)—with a single hyperparameter set and no per-dataset tuning. I prove five formal theorems providing local distinguishability (via the Inverse Function Theorem), global sensitivity (uniform bound $\underline{\sigma} > 0$), finite-sample concentration (Hoeffding), SDP-optimal fusion weights, and a $(1-1/e)$-approximation guarantee for greedy operator selection via monotone submodularity. The paper also introduces a GravityEngine N-body scoring module with a closed-form erf attraction potential that achieves AUC 0.899 on mammography with monotone energy descent. Target venue: **NeurIPS 2026**.

Paper 3: Systemic Risk as Geometry (Adaptive Friction + Navier-Stokes). My third paper—and the centrepiece of the research programme—establishes a geometric framework linking the BSDT deviation operators to the Lyapunov energy of networked agent dynamics. It proves three theorems (equivalence of the detection functional and the physical potential; a spectral phase transition at a critical manifold where constant coupling becomes impossible; and a closed-form optimal damping policy $\gamma^*(E) = E/(E+\theta)$) and validates them across four domains using exclusively real-world data:

- **Banking** (140 quarters of FDIC call-report data, 1990–2024): 6-quarter lead before the 2008 GFC using a threshold trained exclusively on pre-2006 data; AUROC = 0.805 at institution level for the 30 largest US banks. A 24-institution global panel spanning US, UK, EU, Asia, and Africa (using World Bank GFDD, ECB MIR, and BIS bilateral exposure data) achieves AUROC = 0.811 with a 19-quarter GFC lead. Threshold Granger causality passes on the G-SIB panel ($p = 0.043$). The framework also calibrates countercyclical capital buffers: the 2008 GFC consumption-equivalent welfare loss reached 19.57 percentage points; the COVID peak CCyB calibration was 231 basis points.
- **Algorithmic stablecoins** (Terra/Luna UST, May 2022): detection at hour 23, one hour after the initial depeg event, with simulation-to-reality correlation $r = 0.996$. Adaptive friction limits the depeg to 0.28% (versus >99% collapse without intervention).
- **Energy infrastructure** (Texas ERCOT, February 2021): detection at hour 24, a full +72 hours before rolling blackouts begin. Adaptive friction reduces peak capacity loss by 71%. The per-agent decomposition reveals that the cascade origin shifted from frozen wind turbines (the public narrative) to gas supply chain disruption (the structural vulnerability)—a finding invisible to event-study analysis.
- **Fluid dynamics** (3D incompressible Navier-Stokes): reinterpreting γ^* as a state-dependent viscosity yields the **P/D halving principle**—adaptive viscosity halves the enstrophy production-to-dissipation ratio at every Reynolds number tested ($\mathrm{Re} \in [314, 6283]$), confirmed numerically on a pseudo-spectral solver for the Taylor-Green vortex. This constitutes a new conditional regularity criterion connected to the Clay Millennium Prize Problem and identifies three strategies toward closing the gap to unconditional regularity (with explicit disclaimers that it does not solve the Clay problem).

In all three non-financial domains, the dominant BSDT channel is δ_C (camouflage)—the mechanism by which an unstable system conceals its proximity to collapse—demonstrating that the mathematical structure is genuinely universal, not fitted to any single domain. The unifying object is the blind-spot energy functional $E_{\mathrm{BS}} = \delta_C^2 + \delta_G^2 + \delta_A^2 + \delta_T^2$, which serves simultaneously as a fraud-detection diagnostic, a systemic risk alarm, and a viscosity schedule for turbulent fluids. Target journal: **SIAM Journal on Applied Mathematics**.

3. Intellectual property.

On 4 March 2026, I filed a UK patent application with the Intellectual Property Office (Patents Act 1977) titled *"Adaptive Multi-Spectrum Anomaly Detection, Risk-Aware Transaction Processing, and Cross-Domain Stability Control System."* The application contains 25 claims organised into 13 groups covering: (A) BSDT four-channel decomposition and post-hoc correction, (B) UDL full-rank detection guarantees, (C) GravityEngine N-body scoring, (D) Dimension Magnifier nonlinear transform, (E) AMTTP risk-aware blockchain protocol with 4-tier escrow, (F) zkNAF zero-knowledge compliance, (G) cross-chain risk propagation, (H) UI integrity verification (anti-Bybit), (I) adaptive friction for systemic risk and CCyB calibration, (J) state-dependent viscosity for Navier-Stokes, (K) compliance orchestrator with parallel microservices, (L) cross-domain unification via the blind-spot energy functional, and (M) decentralised dispute resolution for ML-flagged transactions.

4. Open-source contributions and tools.

The full codebase—research code, pipeline implementations, client SDKs, smart contracts, and replication packages—is publicly available on GitHub at `github.com/segetii/fintech`. Major open-source components include:

- **TypeScript client SDK** (`@amttp/client-sdk`): 19+ service modules with EIP-712 typed signing, Flashbots MEV protection, HMAC-SHA256 webhook signing, and batch scoring for up to 10,000 transactions.
- **Python client SDK** (`amttp-sdk`): 17 async service modules mirroring the TypeScript API, MIT licensed.
- **MFLS SDK** (`mfls`): complete implementation of the BSDT/MFLS systemic risk engine with BSDT operators, 5 scoring variants, GravityEngine, phase-transition detection, and a FastAPI REST API for systemic risk signal, blind-spot audit, CCyB calibration, and herding monitoring.
- **MetaMask Snap**: browser-based compliance verification for everyday users.
- **GravityEngine toolkit**: N-body simulation for networked agent dynamics with erf-attraction, log-repulsion potentials, spectral radius computation, and trajectory analysis.
- **3D Navier-Stokes solver**: pseudo-spectral solver with BSDT diagnostic suite and adaptive viscosity, including Taylor-Green vortex benchmarks across $\mathrm{Re} \in [314, 6283]$.
- **Complete replication packages** for all empirical results in all three research papers.

5. Why the UK, and why now.

The UK is positioning itself as the global leader in responsible fintech regulation. The FCA's evolving framework for crypto-assets will create substantial demand for compliance infrastructure that operates at the protocol level. AMTTP is, as far as I am aware from surveying the commercial landscape, the only system that provides pre-settlement AML enforcement on blockchain transactions.

The cross-domain reach of my research—from banking regulation to energy grid resilience to pure mathematics—aligns with the UK's strategic priorities in financial stability (Bank of England, PRA), infrastructure resilience (Ofgem, National Grid ESO), and mathematical sciences (EPSRC, the Heilbronn Institute). The Navier-Stokes extension, in particular, connects directly to the UK's world-leading position in applied mathematics and fluid dynamics research.

My immediate plans are: (1) submit the three papers to their target journals (IEEE TKDE, NeurIPS 2026, SIAM J. Applied Math); (2) prepare a short letter for *Nature Communications* summarising the cross-domain universality result for broader scientific visibility; (3) commercialise AMTTP as a compliance-as-a-service product for UK-regulated financial institutions and virtual asset service providers; and (4) build a research-led company at the intersection of machine learning, financial regulation, and mathematical stability theory—headquartered in the UK.

I have done this work alone, without a university laboratory, a GPU cluster, or a research budget. In three years, I have gone from a practical question about missed fraud to a unified mathematical framework that detects collapse across banking systems, stablecoins, power grids, and turbulent fluids—published two papers, filed a patent with 25 claims, deployed 18 smart contracts, trained models on 2.64 million transactions to ROC-AUC 0.9879, and open-sourced the complete stack. What I bring is the ability to move between rigorous mathematical theory and production-grade systems, and the discipline to validate claims before making them. I believe I can contribute meaningfully to the UK's digital economy, scientific reputation, and regulatory leadership, and I am asking for the opportunity to do so.